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ชื่อโครงการ
โซนารี : เส้นทางไร้เสียง
(Sonari : Path Unspoken)
หมวดโปรแกรม เพื่อความบันเทิง
ระดับนักเรียน

รายงานฉบับสมบูรณ์
เสนอต่อ
สำนักงานพัฒนาวิทยาศาสตร์และเทคโนโลยีแห่งชาติกระทรวงการ
อุดมศึกษา วิทยาศาสตร์ วิจัยและนวัตกรรม

ได้รับทุนอุดหนุนโครงการวิจัย พัฒนาและวิศวกรรม
โครงการแข่งขันพัฒนาโปรแกรมคอมพิวเตอร์แห่งประเทศไทย ครั้งที่ 28
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โดย
นางสาวปุณยนุช จิตแกมมณี
นายณัฐวรรธน์ พงศ์นพรัตน์
นายวสุ เลิศธนกิจเจริญ

อาจารย์ที่ปรึกษา นายดวิษ เทพชาตรี
โรงเรียนสาธิตนานาชาติ มหาวิทยาลัยมหิดล
จังหวัด นครปฐม

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ขอขอบพระคุณสำนักงานพัฒนาวิทยาศาสตร์และเทคโนโลยีแห่งชาติ (สวทช.) ที่ได้ให้การสนับสนุนทุนอุดหนุนในโครงการแข่งขันพัฒนาโปรแกรมคอมพิวเตอร์แห่งประเทศไทย ครั้งที่ 28 แก่โครงการ “โซนารี : เส้นทางไร้เสียง (Sonari : Path Unspoken)” ซึ่งการสนับสนุนครั้งนี้มีส่วนสำคัญอย่างยิ่งต่อการพัฒนาและสำเร็จลุล่วงของโครงการ

ผู้พัฒนา

Abstract (บทคัดย่อ)

Thai (ภาษาไทย)

โครงการ โซนารี : เส้นทางไร้เสียง เป็นซอฟต์แวร์เพื่อการเรียนรู้ภาษามือในรูปแบบเกม (Game-Based Educational Software) ที่มุ่งลดอุปสรรคด้านการสื่อสารระหว่างบุคคลที่มีความบกพร่องทางการได้ยินและบุคคลทั่วไป พร้อมทั้งส่งเสริมการเรียนรู้ภาษามือผ่านประสบการณ์การเรียนรู้ที่สนุกและมีส่วนร่วม โดยมุ่งเน้นกลุ่มเด็กและเยาวชนอายุ 6–15 ปี ภายในสภาพแวดล้อมแบบโลกเปิด (Open-World Environment) ที่เอื้อต่อการเรียนรู้และการฝึกฝนภาษามืออย่างต่อเนื่อง

แพลตฟอร์มนี้พัฒนาขึ้นบนกรอบแนวคิดการเรียนรู้ Observe–Understand–Apply (O-U-A) ซึ่งช่วยให้ผู้ใช้สามารถเรียนรู้ฝึกฝน และประยุกต์ใช้ภาษามือผ่านการสำรวจและปฏิสัมพันธ์ภายในเกม ผู้เรียนจะได้รับความรู้จากการสื่อสารกับตัวละครภายในเกม (NPCs) ที่นำเสนอภาษามือผ่านสถานการณ์จำลองที่อ้างอิงจากบริบทการสื่อสารในชีวิตประจำวัน ภาษามือที่เรียนรู้ใหม่จะถูกนำเสนอผ่านระบบ Sign Dictionary แบบบูรณาการ การสาธิตท่าทางผ่านแอนิเมชันของตัวละคร และคำอธิบายบนหน้าจอ เพื่อช่วยให้ผู้เรียนเข้าใจทั้งความหมายและวิธีการใช้งานของภาษามือแต่ละคำอย่างถูกต้อง จากนั้นผู้เรียนสามารถนำความรู้ที่ได้รับไปประยุกต์ใช้ผ่านภารกิจ การแก้ปัญหาเชิงโต้ตอบ กิจกรรมการสำรวจ และการสื่อสารภายในโลกของเกม

เพื่อส่งเสริมการมีส่วนร่วมอย่างต่อเนื่องและเพิ่มโอกาสในการเรียนรู้ซ้ำ แพลตฟอร์มได้ออกแบบระบบความก้าวหน้า (Progression System) ควบคู่กับระบบรางวัล ซึ่งประกอบด้วยเนื้อหาที่สามารถปลดล็อกได้ รางวัลสกุลเงินภายในเกม ความสำเร็จ (Achievements) และตัวเลือกการปรับแต่งประสบการณ์การเล่นตามความต้องการของผู้ใช้ นอกจากนี้ยังมีการประยุกต์ใช้เทคโนโลยี Motion Capture ผ่านเว็บแคมเพื่อสนับสนุนการฝึกภาษามือเชิงปฏิบัติ โดยอาศัยความท้าทาย

ประสบการณ์ใหม่ และระบบรางวัลเป็นแรงจูงใจให้ผู้เรียนเกิดการฝึกฝน และเรียนรู้อย่างต่อเนื่อง

ด้วยการผสมผสานระบบการเรียนรู้ภาษามือเข้ากับรูปแบบการเล่นที่สร้าง การมีส่วนร่วม โครงการนี้มีเป้าหมายเพื่อเพิ่มความสนใจในการเรียนรู้ ภาษามือ ขยายโอกาสในการเข้าถึงการศึกษาด้านภาษามือสำหรับ เยาวชน และมีส่วนช่วยลดอุปสรรคด้านการสื่อสารระหว่างบุคคลที่มีความบกพร่องทางการได้ยินและบุคคลทั่วไปในสังคม

English (ภาษาอังกฤษ)

Sonari: Path Unspoken is educational software with a game format for learning sign language that attempts to bridge communication gaps between the deaf and hearing communities by making the process of learning sign language fun and interactive. The game creates an open-world environment that promotes learning of sign language, especially amongst kids and young adults aged 6-15.

The platform makes use of the Observe, Understand, and Apply (O-U-A) model of learning, which allows the users to learn, practice, and apply sign language in their interactions. The learners acquire knowledge about sign language through interactions with non-player characters (NPCs), who introduce signs through communication scenarios. New signs are introduced to the users via the built-in Sign Dictionary, NPC animations, and sign captions, enabling them to learn the meanings and application of the signs. Finally, the learner applies their knowledge through exploration activities, interactive challenges, quests, and communication tasks.

In order to ensure that users will continue participating and repeatedly be exposed to educational materials, the platform has a system of rewards tied to progression, which includes unlockables, in-game currency rewards, achievements, and personalized options. Webcam-based motion capture is used to ensure sign language practice, whereas new experiences and challenges serve as motivation for learning.

Using sign language learning systems in an immersive game environment, the goal of the project is to enhance engagement in sign language learning, provide more access to sign language education for younger generations, and decrease barriers to communication between the hearing and the deaf communities.

Keywords (คำสำคัญ)

Open-World — A game design approach in which players are free to explore a continuous environment at their own pace, allowing learners to encounter NPCs and challenges naturally through exploration rather than a fixed, linear sequence.

Educational Game-Based learning — A teaching method that uses games to help students learn.

Sign Language Recognition — The use of computers and artificial intelligence to translate sign language gestures into text or spoken words.

Unity — A cross-platform game engine used to build and render the game environment, handling graphics, physics, and user interaction through C# scripting.

Antigravity — A Python-based IDE used during development to write, test, and run the core machine learning and motion capture scripts that power the gesture recognition system.

MediaPipe — A Google framework used to detect and track hand landmarks from webcam input, providing the precise finger and joint position data that the gesture recognition system relies on.

Introduction (บทนำ)

More than 430 million people are estimated by WHO (World Health Organization) to be living with disabling hearing loss worldwide, with the World Federation of the Deaf estimating that 70 million of these people belong to signing deaf communities. That is to say that tens of millions of people use sign language as their main means of communication while the number of hearing people able to reply back in sign language is considerably smaller. As a consequence of this, deaf people and hard-of-hearing individuals are frequently misunderstood, isolated, or even excluded from ordinary interactions just due to the fact that many people do not know how to communicate through sign language. For children and teenagers, who are in their early stages of development of good communication habits, sign language learning is usually presented as something complicated, strange, and unattractive.

It led to the development of the game-based learning platform which is aimed to turn sign language education into something immersive and interactive rather than a repetitive and stress-generating experience. The platform will take children and teenagers through realistic communicational situations which will allow them to learn the signs in a more natural way using contextual approach and repeating the signs actively. The platform called Sonari: Path Unspoken (โซนารี: เส้นทางไร้เสียง) utilizes the technology based on motion capture with the help of a webcam, computer vision and machine learning and open-world game environment in order to educate children at the age from 6 to 15 years old about sign language. The platform

uses the O-U-A (Observe-Understand-Apply) learning framework which consists in the fact that the user will be observing the introduced signs performed by NPCs (non-player characters), will understand the meaning of the signs with the help of Sign Dictionary and captions and will be applying what he/she has learned through different challenges, quests and explorations in the game world. The platform has a well-developed progression and rewarding system which will make the technology feel easy and intuitive and face blurring option in order to make the usage of webcam safer.

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Objectives & Goals (วัตถุประสงค์และเป้าหมาย)

This project aims to address the communication barrier between the deaf community and the rest of the population by creating an interesting and effective way for kids and adolescents aged 6-15 years old to learn sign language. Using webcam motion tracking technology, machine learning, and real-time image processing to track the hand shapes and gestures of the learners, check their correctness, and offer feedback immediately to correct the mistakes made during practice.

The proposed platform is structured on the principles of gaming and learning based on the Observe, Understand, Apply approach. In the process, learners will be able to observe new signs via interaction with non-player characters (NPC), understand their meaning with the help of the built-in Sign Dictionary, and put it into practice through quests, puzzles, and exploring.

Other than imparting language skills, the project has another objective that goes beyond. The introduction of sign language at such an early age is expected to foster awareness of the deaf community as well as inclusion. This way, users will gain confidence when interacting with deaf people beyond the game environment. In essence, the main objective of this project is to enable children to acquire useful sign language skills.

Development Details (รายละเอียดของการพัฒนา)

Storyboard (เนื้อเรื่องย่อ)

Sonari: Path Unspoken follows the journey of a young “outsider”, who is unexpectedly transported from Earth to the fantasy world of Sonari, exploring a fantasy world where communication has gradually been forgotten. Upon the user’s path to return home, the user must learn how to communicate with the locals in Sonari. The user will travel across villages and regions, meeting NPCs from different backgrounds who introduce sign language through contextual interactions and everyday situations. As players complete quests, solve challenges, and practice signs using webcam motion capture, they gradually unlock new areas, rewards, and customization options, creating a progression-driven learning experience that seamlessly integrates education with gameplay.

- Illustrations

1. User Login

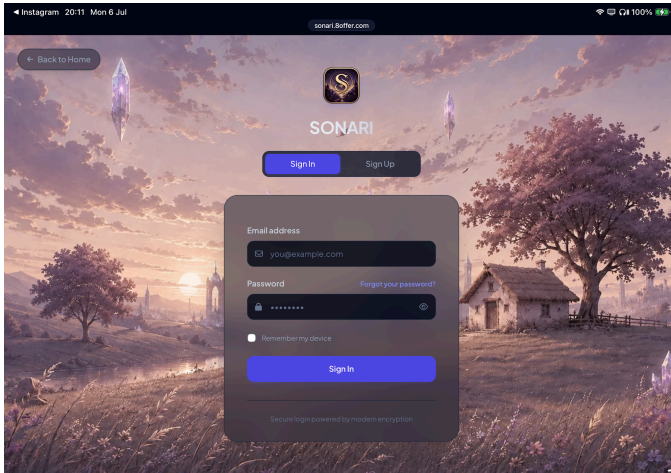


Figure 1.1: Storyboard Illustrations: User Login

The user login system serves as an entry point to the world of Sonari: Path Unspoken. Upon successful account creation, the system will collect and memorize the user's learning progress, unlocked content, achievements and personalized settings, allowing continuous learning experience for multiple sessions. By storing an individual's progress, users are able to resume their progress at any time throughout the platform.

2. NPCs Interaction & Quest Menu



Figure 1.2.1: Storyboard Illustrations: Quest Menu Icon



Figure 1.2.2: Storyboard Illustrations: NPCs Interaction

NPCs interaction acts as the primary method of introducing sign language to users throughout the game. Users are to encounter various NPC archetypes that represent different communication-related scenarios. Through situational conversation with NPCs, and sign demonstrations along with sign-to-text captions, users are introduced to new vocabularies, expressions and mannerisms that relate to the situation's surroundings; this allow users to understand both the meaning of each sign and the situation in which the it should be used. In the first version, NPCs will introduce users to a total of 3 signs. This learning approach encourages users to understand sign language through practical communication, rather than memorization alone.

3. Webcam Practice

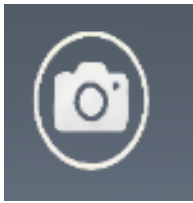


Figure 1.3.1: Storyboard Illustrations: Webcam Icon

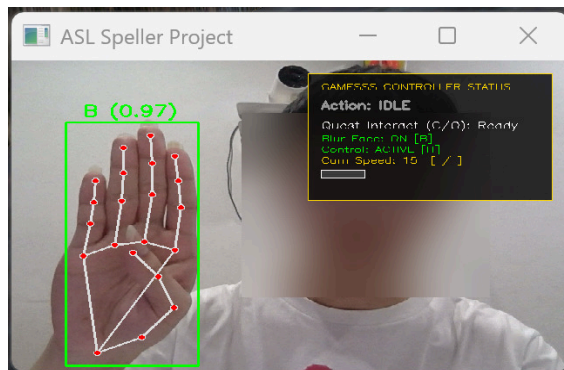


Figure 1.3.2: Storyboard Illustrations: Webcam Practice

The webcam practice module allows users to actively practice acquired sign language through real-time hand gesture practice, using webcam hand motion capture technology. The system will track users' hand movement, and it will identify the signs users have performed. Throughout the game's story, the first version of Sonari: Path Unspoken contains a mandatory practice mechanism in which learners are required to successfully perform a sign gesture five consecutive times before progressing to the next stage of a quest. Through active participation in the learning process, the mechanism aims to support long-term continuity while creating a more engaging and interactive experience.

4. Sign Dictionary

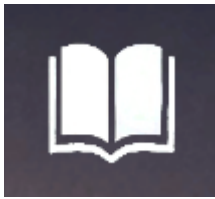


Figure 1.4.1: Storyboard Illustrations: Sign Dictionary Icon

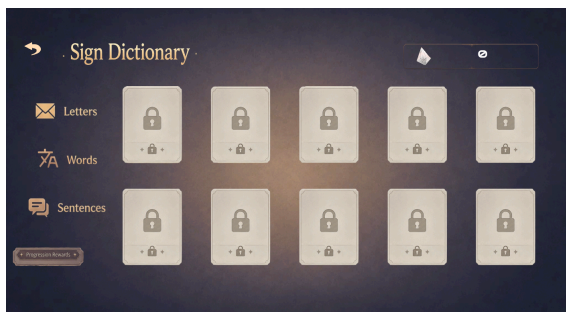


Figure 1.4.2: Storyboard Illustrations: Sign Dictionary Heads-Up Display

The Sign Dictionary functions as a storage where users can review previously acquired signs at any point during the gameplay. There are a total of 3 categories in the Sign Dictionary: letters, words, and sentences. While the first version will only offer the letters category, development of future versions aims to expand the signs' vocabulary and offer users a wider range of entries for users to review. The dictionary also serves as a progress tracker by recording learned signs, allowing users to freely monitor their own learning progress.

5. Reward Progression



Figure 1.5.1: Storyboard Illustrations: Rewards Progression Icon

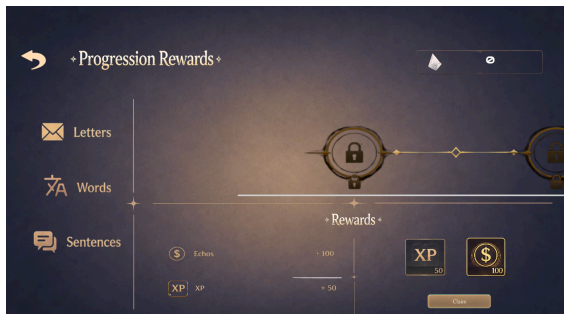


Figure 1.5.2: Storyboard Illustrations: Rewards Progression Heads-Up Display

The Reward Progression system is designed to encourage users' participation by rewarding users' learning achievements throughout the game. Upon learning more signs, players will receive currency and user experience (XP) rewards that correspond to their learning milestones. Rather than functioning solely as a gameplay reward, the Rewards Progression provides positive reinforcement that motivates users to continue exploring, practicing and advancing through the game's educational content.

6. Shop & Currency

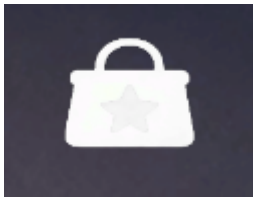


Figure 1.6.1: Storyboard Illustrations: Shop Icon

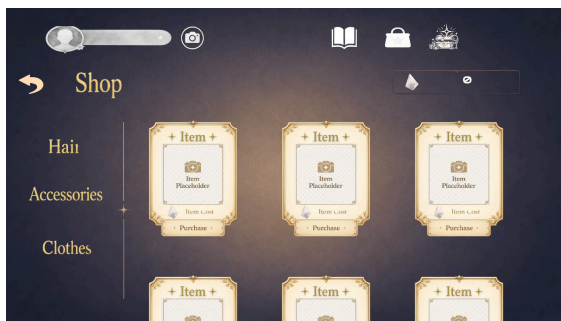


Figure 1.6.2: Storyboard Illustrations: Shop Heads-Up Display

The Shop & Currency system introduces users an additional motivational factor by rewarding users with the in-game currency, called Echoes, for completing activities, including quest completion, NPC interaction, collecting signs and exploration. Users may exchange their Echoes for avatar-customization items, such as cosmetics, clothes, and accessories. By linking rewards directly to learning activities, the system encourages repeated participation for users, while providing incentives for continuous learning with the platform's educational content.

Related Theories (ทฤษฎีหลักการ)

The development of Sonari: Path Unspoken is mainly guided by learning and motivational theories that support the design of the software's learning framework and gameplay systems. The following theories form the conceptual foundation of the proposed software.

1. Kolb's Experiential Learning Theory

Kolb's Experiential Learning Model, developed by psychologist David A. Kolb in 1976, emphasizes the process of learning through experience. This model is rooted in the belief that effective learning occurs through a cyclical four-step process: Concrete Experience, Reflective Observation, Abstract Conceptualization, and Active Experimentation. The Observe, Understand, and Apply learning framework is mainly derived from Kolb Experiential Learning Theory. Similar to Kolb's Learning Cycle, users will encounter sign language through in-game experiences—such as NPCs interaction and challenges. Built-in features—such as Sign Dictionary—allows users to further develop their understanding of sign language through repeated reviews and exposure. Ultimately, the webcam practice features allow users to apply their attained knowledge of sign languages directly. These features, together, promote continuous learning through repeated encounter, understanding and practice.

2. Situated Learning Theory

Situated learning theory (SLT), first presented by Jean Lave and Etienne Wenger (1991), explains the process and development of learning when individuals have the opportunity to participate in a community of practice. The software applies this theory into practice by introducing users to NPCs' interaction within contextualized content. By applying sign language learning in the gameplay, users are encouraged to both understand the meaning of individual signs and the appropriate situations where they should be used. Sonari : Path Unspoken introduces different types of NPCs, each of them represents different everyday-communication settings. Each type of NPCs will introduce users to different types of vocabulary, mannerism and conversations, allowing users to understand and apply sign language appropriately in different real-life communication situations.

3. Gamification Theory

Gamification Theory involves adding game-like features to learning, with the goal of motivating students and increasing engagement. Elements from games like earning points, collecting badges, competing on leaderboards, and taking on challenges are used in education. This makes lessons more interactive and immersive than traditional ways of teaching. Sonari : Path Unspoken applied this theory by integrating game elements into the learning process, through game features such as reward progression, achievements, and a currency system that intends to increase user engagement. In-game currency, called “Echoes”, is awarded to users upon quest, achievements, and

puzzle completion; this currency allows users to exchange for avatar customization items. Through the use of this theory, systems are designed to support long-term learning by encouraging users to enjoy learning sign language through gamified educational features.

Technology or Techniques Used (เทคนิคหรือเทคโนโลยีที่ใช้)

Computer vision

- Used to detect and track the user's hands with precision as they move through the learning environment. This allows the system to interpret gestures in real time and respond smoothly to each action.

Motion capture

- Used to record the position and movement of the hands as the user performs signs. This creates a more natural and responsive learning experience by closely following each gesture.

Sign Language Recognition

- The use of computers and artificial intelligence to translate sign language gestures into text or spoken words.

Machine Learning

- A subset of artificial intelligence where algorithms analyze large datasets to recognize patterns.

Gamification

- Used through reward systems and interactive challenges that make learning more engaging. By turning practice into

play, it encourages users to stay motivated and continue improving their sign language skills.

Python

- The hand gesture recognition component of the project was created using Python. It was chosen because Python can be used for machine learning, computer vision, and motion tracking, which means that it allows developing scripts that can work with webcam footage and analyze hand gestures in real time. In addition, Python makes it easier to link different vision and AI components needed for gesture recognition.

C#

- The C# programming language was used to create the scripts that control the game and its various interactive elements within the Unity engine. C# scripts were needed to define the game behavior, including interactions between the user character and NPCs, quests, UI, and other elements of the game within the Unity engine.

HTML

- HTML language was used to create the web-based parts of the application. The role of HTML is to provide the basis for the structure of the pages and includes interface elements, such as buttons and text.

PHP

- PHP programming language has been used for developing the login webpage as it provides the capability to develop the secure server side processing along with the integration of HTML to provide a functioning webpage. Here, HTML has been used for designing the form while PHP has been used for processing the information provided by the user.

MediaPipe (Hand Tracking)

- The MediaPipe library was used to detect the hand of the user through the webcam and track 21 landmarks on its joints in 3D coordinates. It was possible due to the capabilities of this library to detect and track the movements of the hands of the user, which enabled the analysis of sign language gestures in real-time. MediaPipe has played an important role in the gesture recognition process since it helped in detecting landmarks of the hand.

NumPy (AI Math and Neural Network)

- The library NumPy has been applied to compute the coordinates of hand landmarks and do the necessary mathematical computations for the neural network model. It was essential for processing the landmarks detected by MediaPipe and helped the LiteMLP model to determine which ASL letter is spelled by the user.

OpenCV (Camera and Display)

- The OpenCV library was used to open the webcam, receive live video from it, and display the stream on the screen. Additionally, this library was responsible for the

implementation of the face-blur feature that was implemented to ensure user privacy while using the webcam. Finally, OpenCV was able to work with video frames to process them in real-time so that the gesture recognition module could recognize hand gestures.

Pydirectinput (Game Keyboard Controls)

- Pydirectinput library was responsible for generating keyboard inputs, like keys W, A, S, D, E and Space directly into the Unity game engine. This allowed the system to move the character in the game world based on the recognized gestures of the player. It was very useful for interaction of the hand recognition system with gameplay mechanics.

Pyautogui (Camera Mouse Control)

- Pyautogui was responsible for controlling the virtual mouse pointer and therefore rotating and dragging the camera in the game with the help of the player's camera.

Development Approach (เครื่องมือที่ใช้พัฒนา)

1. Antigravity

The Integrated Development Environment (IDE) based on the Python programming language used for coding and executing machine learning, computer vision and motion capture script code constituting the main components of the gesture recognition system.

- Applications: Machine learning model execution, MediaPipe library integration, gesture recognition pipeline running in the webcam environment.

2. Unity

A cross-platform gaming engine used for creating and rendering the game environment, including open-world map, NPC interactions, UI elements, challenge system and advancement mechanics, all implemented via C# scripts.

- Applications: Game world creation, NPC behavior implementation, quest/reward system creation, avatar customizations.

3. FileZilla

FTP client used for transferring files between local development environment and the web server to facilitate remote updating and maintenance of the web-based platform.

- Applications: File management of the website, deploying updates of HTML and JavaScript to the server.

4. Notepad

A basic text editor used in the early phase of development for writing and editing the raw HTML code required for the structure of the platform website.

- Applications: Creating the structure of webpages and editing HTML code.

5. Visual Studio Code

An extensive code editor which was used for writing and managing the script logic of the game.

- Applications: Scripting the game logic and web interfaces.

6. Canva

Graphic design tool used to design the visual and UI elements of the game, such as the mockups, icons, promotional materials, and in-game visuals.

- Applications: UI design, designing game visual elements, creating posters and presentations.

7. GitHub

Version control and coding platform used to refer to open-source implementations of existing approaches in order to implement gesture recognition and game integration.

- Applications: Reference for implementing ML model, pipeline for gesture recognition, and Unity scripts.

8. ChatGPT

Artificial intelligence that helps create visuals for the game such as UI concepts and environment design.

- Applications: Visuals for game environment, game UI, game website, and non-playable characters concept.

9. Copilot

AI assistant developed by Microsoft.

- Applications: Help debugging unity script.

10. Google Gemini

AI assistant that helps with coding, offers suggestions and solutions when faced with some technical problems in the project.

- Applications: Python ML scripts, C# Unity script, and others.

11. WinRAR

A file compression tool used to compress project files and package the software into a .exe format for easier distribution and installation.

- Applications: Compressing game files, creating .exe files, and preparing the software for sharing or deployment.

12. Suno AI

AI-based music generation tool used to create background music for the game, helping enhance the atmosphere and player experience.

- Applications: Creating game BGM, improving immersion, and supporting the overall game presentation.

13. Capcut

A video editing tool used to edit and combine project videos, promotional clips, and presentation materials for the game.

- Applications: Editing promotional videos, gameplay clips, demonstration videos, and presentation content.

Software Specifications (รายละเอียดโปรแกรม)

Input & Output Specifications

Input-output specification describes how the software gets information from users and reacts in various modules. In this case, the software integrates education, gaming, and AI-powered tracking of body motions to create a comprehensive interactive sign language system for kids and teenagers. This way, a clear description of what goes in and what is generated helps all modules of the login system, the educational part, the webcam interactions, and the rewards systems interact flawlessly.

1. Login Specification

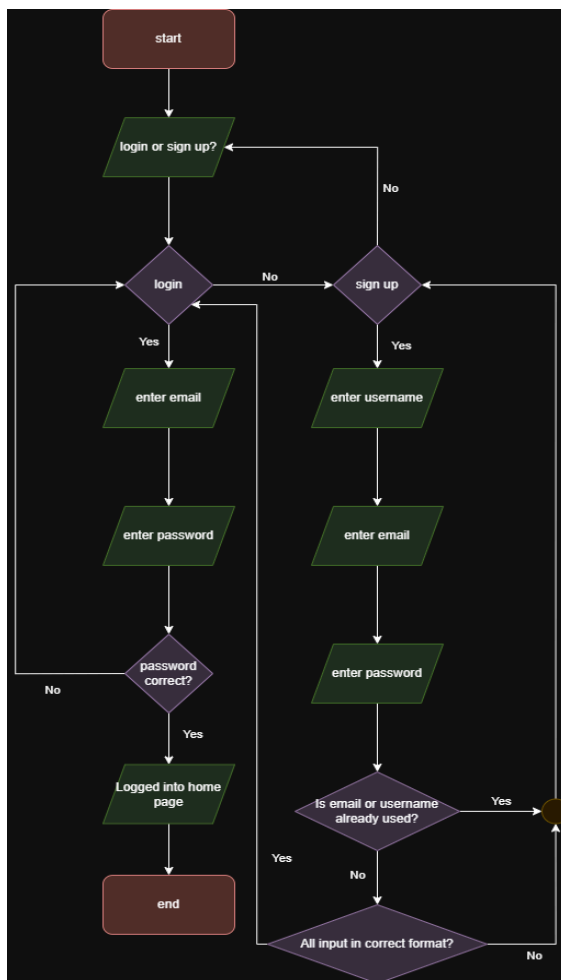


Figure 2.1: Software Specification: Login Specification

This flowchart shows how account access works in Sonari. When users open the platform, they are asked to either log in or sign up. Logging in requires a registered email and password, and if the password is incorrect, the user must enter the information again. Signing up requires a username, email, and password, all of which must follow the correct format and not already be linked to another account. Once the information is verified, the user is authenticated and taken to the home page.

2. Learning System Specification

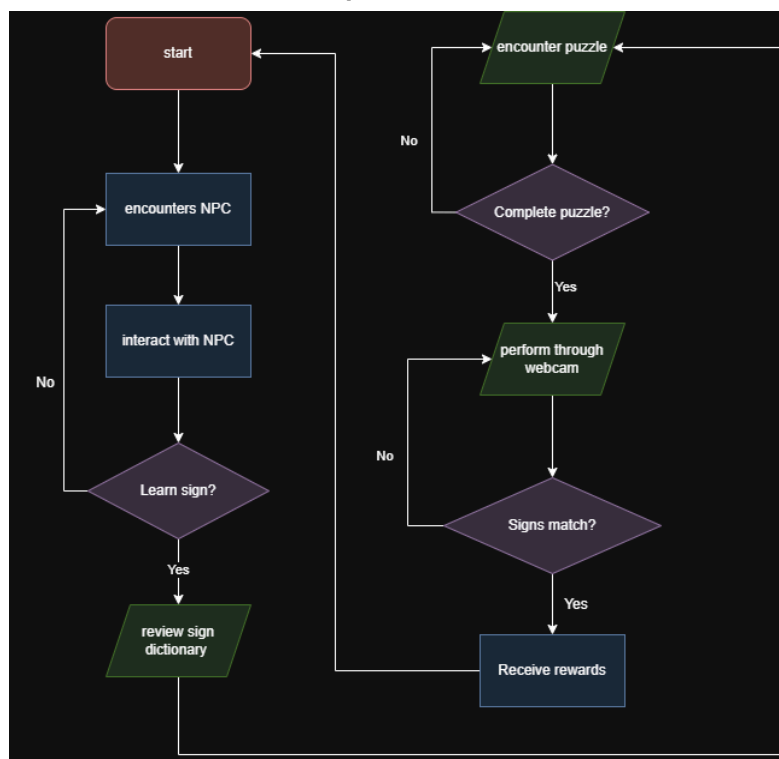


Figure 2.2: Software Specification: Learning System Specification

This flowchart explains the main learning cycle in the game. After entering the game world, users meet an NPC and interact with it to begin a sign language lesson. If they choose not to learn that sign, they return to find another NPC. If they accept, they review the sign in the Sign Dictionary and then complete a puzzle that checks their understanding. If they fail, they retry the puzzle. After completing it, they perform the sign using the webcam, where the machine learning model evaluates their gesture. If the sign is not correct, they practice and try again until the system recognizes it successfully, after which rewards are given.

3. Rewards System Specification

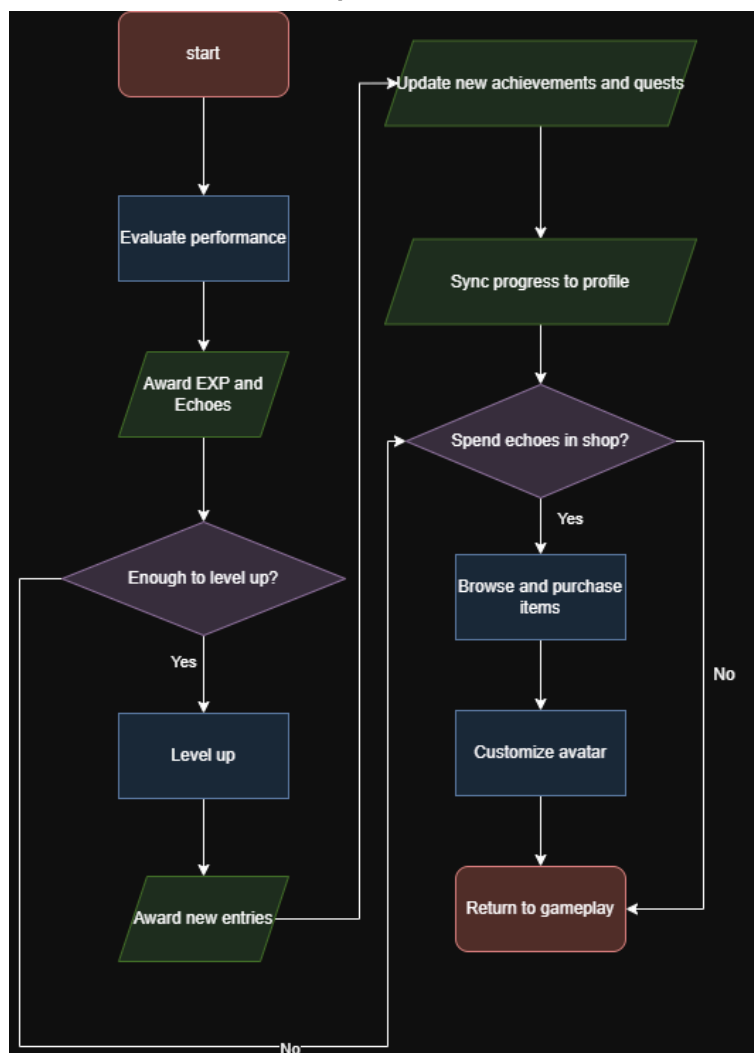


Figure 2.3: Software Specification: Rewards System Specification

This flowchart describes how progress is recorded and rewarded after each learning cycle. The system first checks the user's performance before awarding EXP and Echoes, which is the in-game currency. If the user has enough EXP, they level up and unlock new dictionary entries, achievements, and quests. Even if no level up occurs, the user's progress is still saved to their profile. After that, users can spend Echoes in the shop to buy cosmetic items for avatar customization before returning to the game.

Scope & Limitation (ขอบเขตและข้อจำกัด)

Scope (ขอบเขต)

Sonari: Path Unspoken is a gaming platform that makes the learning of sign languages easy and fun for children and teenagers ranging from ages 6 to 15. The platform employs technologies like webcam motion capture, machine learning, and the use of an open world game environment to take learners through the Observe-Understand-Apply approach of learning. Sonari is meant for absolute beginners in sign languages and is based on learning static American Sign Language alphabet signs with one hand at the moment.

Limitation (ข้อจำกัด)

The project limitation section outlines the current technical and development constraints of the prototype.

- At the moment, the prototype has no means of recognizing motions, and, thus, cannot detect hand signs which require motion. Consequently, motion signs like J and Z cannot be incorporated in the program at the moment because they are dependent on motion and not on a particular hand shape. It would appear then that motion signs will be difficult to incorporate at the moment.

- In addition to the lack of implementation of certain aspects of the game, its incomplete state is a result of the short development period. In particular, the progress of work on this project has been slowed down by the need to learn new tools used in the development process (for example, moving from using JavaScript to C# in Unity). Thus, the current version of the prototype can be regarded as proof of concept rather than an actual game.
- Now the team has limited possibilities for access to UI elements and assets of the game (icons, character models, environments, etc.). That is why some parts of the prototype include only placeholders or extremely simplified graphics in order to show the idea of future development.
- It was not possible to animate the hand gestures of the NPC characters as this would require a much more sophisticated hand model along with a much better animated version of the game which was not yet feasible during the development phase.
- To use the gesture recognition system, each handshape in ASL must be done with utmost precision since it is designed to identify a certain set of hand shapes and landmarks. Small variations in finger positioning, angle, and direction can lead to the inability of the system to accurately read or recognize signs. This can make it difficult for young users to achieve precise results.

- The current prototype does not provide adequate support for screens with different aspect ratios since the entire interface was designed and tested on a particular display size.

Target Group Users (กลุ่มผู้ใช้โปรแกรม)

This project is primarily intended for children and teenagers aged between 6 years to 15 years, an age group selected due to its importance in terms of the language acquisition process and social development. This is the age at which individuals are more receptive to developing new communication skills and are forming their perceptions of inclusiveness towards people with disabilities.

Program Testing Results (ผลของการทดสอบโปรแกรม)

- **Objectives**

The program testing was conducted to evaluate the functionality, usability, and educational effectiveness of Sonari: Path Unspoken. The evaluation focused on verifying feature performance, ensuring the functionality and accuracy of core game elements, and collecting user feedback regarding the overall learning experience.

- **Participants**

Participants	Value
Number of Participants	17
Age Range	10-17 years
Time Taken to Complete the Game	approximately 10-30 minutes

- **Functional Testing**

Features	Tested Participants	Results
Login & Sign up	17	All participants successfully logged in or created an account.
Opening Scene	17	All participants successfully viewed the one-time opening scene.
NPCs Interaction	17	All participants completed the interaction with NPCs.

Quest Menu were	17	All participants able to view the Quest Menu freely.
--------------------	----	--

Character Movement participants were	17	All able to control the character's movement freely.
---	----	---

Webcam Practice	17	All participants successfully completed Webcam Practice.
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Sign Dictionary were	17	All participants able to access and review learned signs.
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Rewards Progression were	17	All participants able to access and claim the rewards.
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Shop & Currency were	17	All participants able to view their obtained Echoes and browse for shop items.
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- **User Evaluation**

Participant's Game Review

Q1: What is your age?

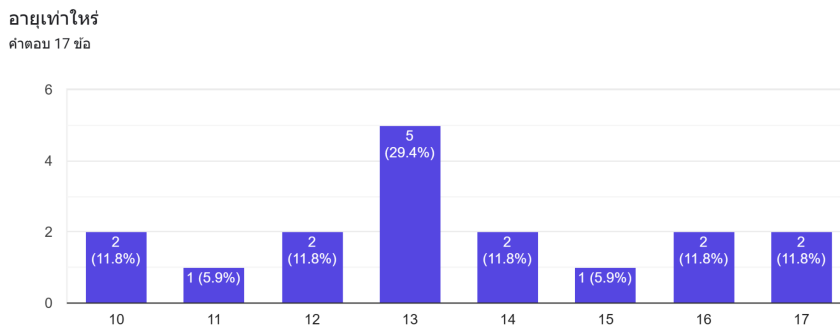


Figure 3.1.1: Program Testing Results: Game Evaluation: Age Range

According to the data collected, the majority of participants, approximately 70%, is in adolescent years, while the rest of participants are young children.

Q2: The game helped me learn sign language.

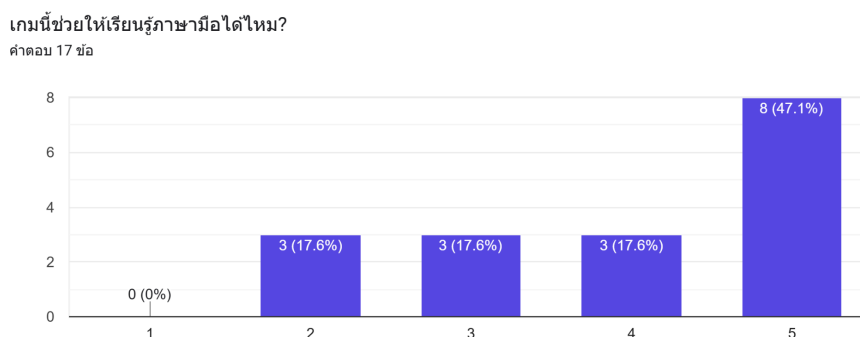


Figure 3.1.2: Program Testing Results: Game Evaluation: Learning Effectiveness

Approximately 65% of participants selected the score of 4-5 for learning effectiveness after completing the first version of

Sonari: Path Unspoken. Although the remaining 35% of participants rated the learning effectiveness between 2 and 3, the results suggest that effectiveness varied among participants, while the majority still evaluated the software as highly effective.

Q3: The webcam and sign gesture practice were easy to use.

ใช้กล้องและท่าทำมือได้ง่ายไหม?
คำตอบ 17 ข้อ

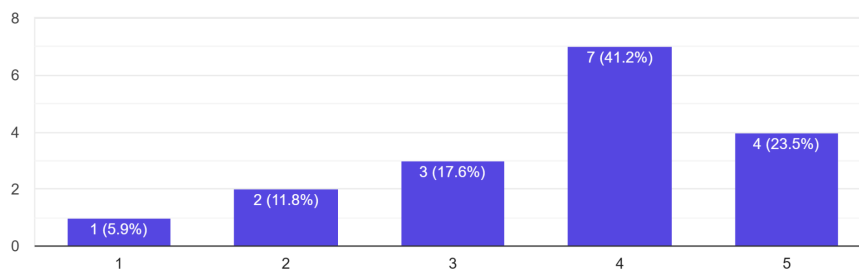


Figure 3.1.3: Program Testing Results: Game Evaluation: Webcam Feature Accessibility

The majority of participants evaluated the webcam's accessibility as high, with approximately 65% of participants scored the accessibility of 4 or above. However, the remaining participants find the features quite challenging, suggesting possible future improvements for an improved user experience.

Q4: I liked the game's visuals, character designs, and overall art style.

ชอบภาพ ตัวละคร และการออกแบบเกมไหม?
คำตอบ 17 ข้อ

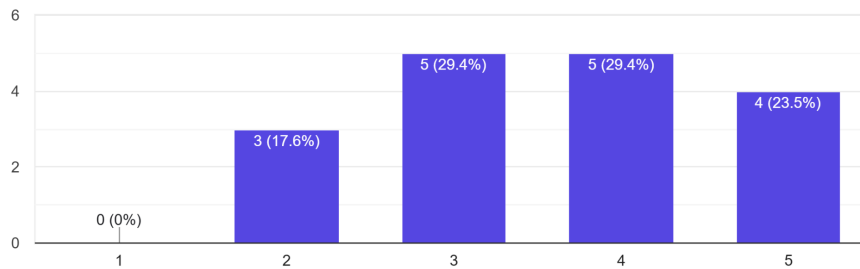


Figure 3.1.4: Program Testing Results: Game Evaluation: Overall Design

Approximately 50% of participants scored the overall design of Sonari: Path Unspoken positively, with a score of 4 or above. While, the remaining 50% of participants are not highly satisfied with the game's design. This evaluation suggests future improvements of the visuals, character and overall designs.

Q5: I found the game enjoyable to play.

เกมนี้สนุกไหม?
คำตอบ 17 ข้อ

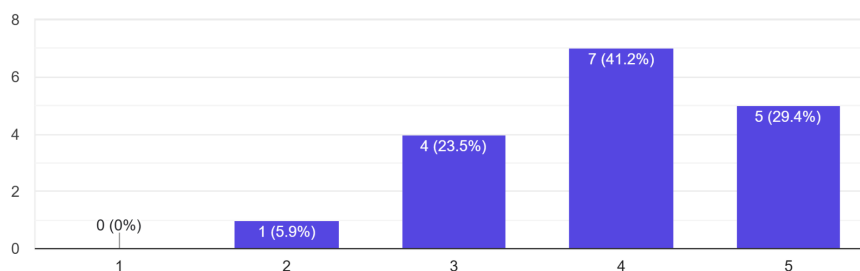


Figure 3.1.5: Program Testing Results: Game Evaluation: Overall Satisfaction

The majority of participants scored positively on overall satisfaction , with 70% of participants providing a high score of 4

or above. Although, remaining participants show slight dissatisfaction with the current game progress.

Participant's Website Review

Q1: What is your age?

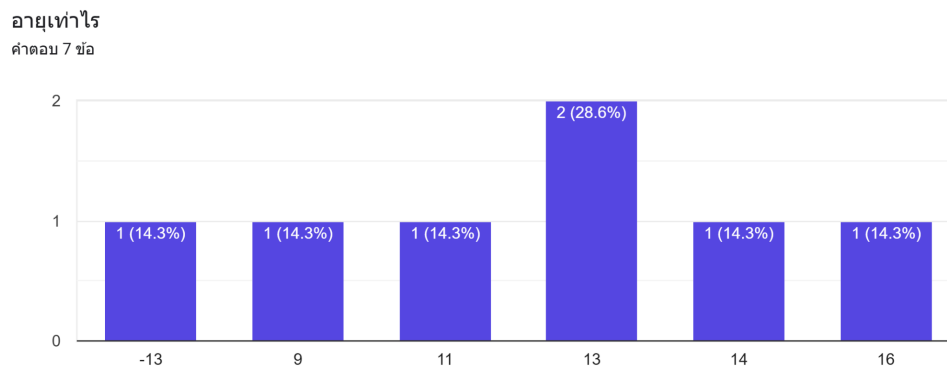


Figure 3.2.1: Program Testing Results: Website Evaluation: Age Range

The majority of the participants vary from each age with them being adolescent while 14.3% of the participants are young children.

Q2: The website is easy to understand.

เว็บไซต์เข้าใจง่ายมั๊ย
คำตอบ 7 ข้อ

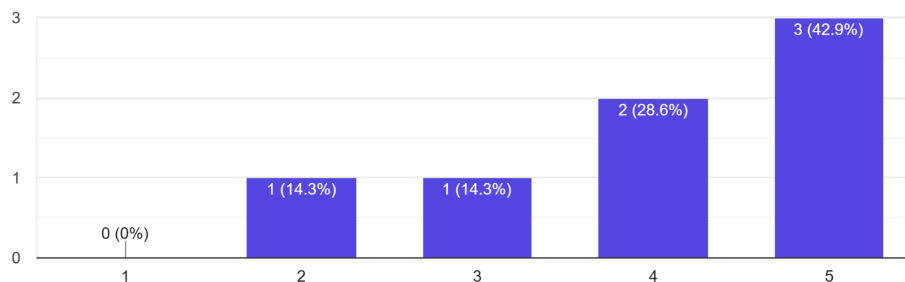


Figure 3.2.2: Program Testing Results: Website Evaluation: Website Comprehension

Approximately 43% of the participants chose the maximum possible score for the website's accessibility to learn something. While some had difficulties, the majority assessed the website as effective.

Results Conclusion & Limitations

It is evident from the findings of the tests that the core objectives of Sonari: Path Unspoken have been effectively met as an educational software prototype through the use of games by the successful working of all its implemented features and positive feedback from the users regarding its learning abilities and the quality of gaming experience. Nonetheless, the present assessment has been carried out on a small number of 17 participants, and the prototype primarily emphasizes the implementation of the core learning elements, which implies limited content and visuals.

Problems & Obstacles (ปัญหาและอุปสรรค)

In this segment, we learn about the challenges encountered in the process of development. Among the challenges encountered were the slowness of gestures' recognition, the alteration of programming tools and also performance challenges that arose as a result of privacy features.

- The team initially used YOLO for hand gestures recognition; however, it proved to be rather slow and inaccurate for real-time detection of hand gestures, it also varies on the lighting, background, and etc. In order to fix that problem, they used MediaPipe which performed way better than YOLO would and was faster and more precise.
- Another issue occurred in Unity – they were supposed to code using JavaScript, however, ended up learning and coding in C# which resulted in an increased development time since they had to change the existing code.
- The web version encountered a problem because the game file was too large, which caused an out-of-memory issue during use. As a result, the project was temporarily changed to an app-based version first, while the team continues looking for a suitable solution to fix the problem.

Future Development & Application (แนวทางในการพัฒนาและประยุกต์ใช้ในขั้นต่อไป)

Explains how the project may grow in later versions. It focuses on improving graphics, interaction quality, storyline depth, and motivational features such as seasonal events and daily rewards to support long-term learning.

- Further iterations of the project should be devoted to improving the visual aspect and interaction of the game so that it seems more completed and immersive. The process involves refining characters, environments, interface, and making webcam hand interactions more responsive. Graphics enhancement and improved interaction design should make the experience more game-like than educational.
- Another possible improvement is creating a story for the game that will provide users with a feeling of achievement and purpose. A well-developed storyline will make interactions with non-player characters and challenges more rewarding because every learned sign will be relevant to the game universe. This will ensure that users are involved in the gameplay for a long period of time.

- Moreover, it is possible to introduce some timed rewards such as holidays events, daily quests or special challenges to ensure regular use of the game. These aspects are aimed at encouraging people to come back to the project to continue learning sign language. Short-term rewards will improve long-term memory.

Conclusion & Recommendation (ข้อสรุปและข้อเสนอแนะ)

Conclusion

Acknowledging the importance of communication in people's lives, especially in communication between deaf and hearing people, the creation of the game Sonari: Path Unspoken was seen not only as entertainment but also as an opportunity to create a learning environment for children and teenagers that will allow them to learn sign language in a fun and interactive way. Thanks to the implementation of Observe-Understand-Apply model of the learning process, exploration in the open world, interactions with NPCs, the use of Sign Dictionary, webcam training of gestures and rewards for progress made, this project makes the process of learning sign language more interesting and motivates people to learn it regularly. With the help of computer vision technologies, machine learning and motion capture used in game development, the platform makes it possible to gain basic sign language skills while also raising awareness, promoting inclusion and encouraging confidence in communicating with the deaf community.

Recommendations

In order to better Sonari: Path Unspoken in future iterations of the project, the following changes are recommended. First of all, it is necessary to expand the database of motion signs and include more complicated hand gestures into the game in order to make the learning process more realistic and comprehensive. It would also be useful to refine the interface of the project in order to achieve its optimal performance on different aspect ratios of screens. Besides that, it is recommended to make the system of recognition of the users' gestures more accurate in order to detect even the slightest gestures and to upgrade visual assets, animations of NPCs and the whole project in order to make it more professional and engaging. Also, it would be helpful to test the project with children of the target age group in order to get more feedback.

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Contact Information (ข้อมูลติดต่อของผู้พัฒนาและที่ปรึกษา)

หัวหน้าโครงการ: Nuttawat Pongnopparat (นายณัฐวรรธน์ พงศ์
นพรัตน์)

Email: noon.nuttawat@gmail.com

Phone Number: 090-329-2528

ผู้ร่วมโครงการ: Phunyanuch Chitgammanee (นางสาวปุณยนุช จิต
แกมมณี)

E-mail: phunyanuchchitgammanee@gmail.com

Phone Number: 080-270-0297

ผู้ร่วมโครงการ: Wasu Lertanakitcharoen (นายวสุ เลิศธนกิจเจริญ)

E-mail: zenwasu18@gmail.com

Phone Number: 097-045-5845

อาจารย์ที่ปรึกษา: Dawit Thepchatree (นายดวิษ เทพชาตรี)

E-mail: dawit.the@mahidol.ac.th

Phone Number: 080-214-3555

Appendix (ภาคผนวก)

Detailed Installation Manual (คู่มือการติดตั้งอย่างละเอียด)

1. Go to the website sonari.8offer.com using your web browser.
2. Sign in to your account. If you do not already have one, create a new account by signing up first. You can't download without signing in.
3. After logging in, click the **Download** button on the website. The file will load automatically.



4. Open download file, then double-click **sonari_installer.exe** to start the installation process.



5. Once the installation is complete, open the **Sonari** game from the desktop to begin playing.



Detailed User Manual (คู่มือการใช้งานอย่างละเอียด)

Introduction

This user manual provides guidance on how to use Sonari: Path Unspoken. It explains the basic features of the platform, navigation, and gameplay mechanics to help users with their learning journey.

- **System requirements**

Requirements	Specifications
1. Webcam	required for gameplay
2. Internet	required for installation
3. Windows Platform	required
4. Supported Browser	required for installation

- **Gameplay Guidelines**

1. Account Registration & Login

This feature allows users to create and access their personal account and continue their learning process. Existing users may log in to their accounts using their registered credentials, while new users may create an account by selecting the Sign Up option.

Steps:

1. Launch the official website through a supported browser.
2. Select Login or Sign Up.
3. Enter the desired account's credentials, for existing users, or required account information, for new users.
4. Click Login to enter the game.

2. Opening scene

Upon the user's initial login, users will be introduced to the story of Sonari: Path Unspoken through an opening cutscene. The introduction is prototypical, focusing on providing background information about the user's role as a traveler, unconsciously transported from Earth to Sonari, and the game's overall objective of finding a way back to Earth.

Steps:

1. Watch the opening cutscene.
2. Read the story narration.
3. Proceed to the first quest after the introduction.

3. World Exploration & Character Controls

Users are allowed to explore the world of Sonari by controlling the main character's movement, and navigate through different settings. Exploration allows users to uncover new areas, NPCs, and progress through the story.

Controls:

Action	Control
Movement	W A S D / SpaceBar / Shift or Hand Controls
● Forward	W or 
● Backward	S or 
● Left	A or 

- Right

D or



- Jump
- Sprint

SpaceBar or
Shift



Camera Movement

Mouse or Hand Controls:

- Switch Camera Perspective



Q

NPCs Interaction

E or



- Continue NPCs Dialogue

SpaceBar or



Webcam Controls

B / H / [/]

- Blur Face Feature On/Off
- Terminate Hand Controls
- Decrease Camera Speed
- Increase Camera Speed

B
H
[
]

Steps:

1. Use the character movement controls to navigate through the game world.
2. Explore different settings and locations, while searching for NPCs.
3. Approach the NPCs as guided by the quest.
4. Press the NPC Interaction key when prompted.
5. Continue to explore, following the quest's guidelines.

4. NPC Interaction & Quests

NPCs will be placed across the game world. Each NPC will introduce signs through contextual conversations, helping users to develop a better understanding of situational context through different communication scenarios. By clicking the Quest Menu, users are able to find the quest's description and guidelines for the next part of the story. Above the user's character, there will be an arrow which guides users to their next destination. After learning a sign, users will receive a required quest to practice the learned signs.

Steps:

1. Explore the game.
2. Check the current quest's objectives in the Quest Menu.
3. Follow the arrow directions.
4. Approach the NPC as stated in the Quest Menu.

5. Interact with the NPC to begin the conversation.
6. Learn the presented signs.
7. Follow through the next steps in the quest.

5. Webcam Practice

The Webcam Practice feature allows users to practice newly learned signs through active participation. Before progressing to the next part of the quest, users are required to perform the signs gesture five consecutive times. The Webcam Practice window will appear at the corner of the game screen; its size will be adjustable by the user. Additionally, for optimal accuracy, ensure that proper lighting and only one user is visible within the camera frame.

Step:

1. Allow camera access.
2. Position one hand within the camera frame.
3. Perform the sign gesture according to the quest's instructions.
4. Repeat the gesture five times.
5. Continue to the next part of the quest.

6. Sign Dictionary

The Sign Dictionary serves as a user's personal learning review space, where users can review previously acquired signs at any point during their learning journey. Once users learn a new sign,

an entry containing the sign's meaning and image will be visible, allowing users to review the acquired signs that correspond to their current progress.

Steps:

1. Open Sign Dictionary by clicking its icon on the top navigation bar.
2. Browse for learned signs.
3. Review the hand gesture of each sign.
4. Return to the game by clicking the exit button on the top left corner.

7. Rewards Progression

Users will receive rewards after completing quests, webcam practice, and unlocking a certain amount of signs. The Rewards Progression will track the user's learned signs and unlock each level's reward after a certain amount of signs are unlocked. The rewards include Echoes, the game's currency, and user XP.

Steps:

1. Complete the assigned quest or challenge.
2. Open Rewards Progression by clicking its icon on the top navigation bar.
3. Receive level's reward by clicking the "claim" button.
4. Return to the game by clicking the exit button on the top left corner.

8. Shop & Currency

Throughout the game, users will earn Echoes, the game's currency. Echoes will be earned by completing quests, webcam practice, and claiming rewards in Reward Progression. Echoes can be exchanged for avatar customization items, including cosmetics, clothes, and accessories. Customization items within the shop are currently represented by a placeholder, while purchased items will become available for customization, and will be expanded and improved in future developments.

Steps:

1. Open the Shop by clicking its icon on the top navigation bar.
2. Check the user's currently owned Echoes, located on the top right corner.
3. Browse for available items.
4. Return to the game by clicking the exit button on the top left corner.

- **Troubleshooting**

If users encounter technical issues while using the software, users should first verify that a stable internet connection, granted webcam access, and a well-lit environment are all available.

Issue	Solution
Unable to log in	Verify login credentials and try again.
Hand motion capture is inaccurate.	Use the webcam in a well-lit environment and ensure that the hand is positioned within the camera frame.
The game loads slowly.	Check the PC's compatibility or restart the game.
Character Movement does not respond to	Perform the gestures slowly and clearly, while ensuring

Hand Controls.

the hands are within the camera's field.

- **Frequently Asked Questions (FAQs)**

Q1: Do I need a webcam to play?

ANS: A webcam is a required part of the quest, to help enforce user's practice. However, other gameplay features, such as exploration and NPCs interaction can still perform without using the webcam.

Q2: Can I replay signs that I've already learned?

ANS: Yes, all previously learned signs can be reviewed in the Sign Dictionary at any time.

Q3: Why isn't my hand gesture being recognized?

ANS: The problem may occur due to hand visibility within the camera frame or an incorrect angle of the hand gesture. Ensure that both elements are consistent, while performing the signs clearly and slowly.

Q4: Why can't I continue to the next quest?

ANS: Some quests will require the users to complete the mandatory webcam practice session before the next part of the story is unlocked.

Q5: Why are there only a few signs and quests available?

ANS: The current version of Sonari: Path Unspoken focuses on the software's core learning loop and gameplay mechanics.

Additional signs, quests, and features are expected to be implemented in future developments.

Q6: How do I earn Echoes?

ANS: Echoes are earned through quest completion, webcam practice completion and progressing through the game.

Disclaimer (ข้อตกลงในการใช้ซอฟต์แวร์)

ข้อตกลงในการใช้ซอฟต์แวร์

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License Agreement

This software is a work developed by Nuttawat Pongnopparat from MUIDS under the provision of Dawit Thepchatree name under Sonari: Path Unspoken, which has been supported by the National Science and Technology Development Agency (NSTDA), in order to encourage pupils and students to learn and practice their skills in developing software. Therefore, the intellectual property of this software shall belong to the developer and the developer gives NSTDA a permission to distribute this software as an "as is" and non-modified software for a temporary and non-exclusive use

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Promotional Poster (โปสเตอร์สำหรับการเผยแพร่)



การแข่งขัน
พัฒนาโปรแกรมคอมพิวเตอร์แห่งประเทศไทย
The 28th National Software Contest: NSC 2026



Sonari: Path Unspoken

(โซนารี: เส้นทางไร้เสียง) — AI-powered Sign Language Game-Based Learning Platform

Introduction

The WHO estimates that over **430 million people** worldwide live with disabling hearing loss.

In other words, **tens of millions** rely on sign language, while far **fewer hearing people** can communicate in it.

Sonari: Path Unspoken is a game-based educational software that promotes sign language learning through interactive gameplay, reducing communication barriers between deaf and hearing communities.

Objectives

- Reduce communication barriers between the deaf community and the wider public.
- Teach sign language to children and adolescents aged 6–15 in an engaging way.
- Use webcam motion tracking, machine learning, and real-time image processing to **detect hand signs**.
- Check gesture accuracy and give immediate feedback to **support practice and correction**.

Key features

Quest Menu
: Shows learning goals and tasks.



Sign Dictionary
: Reviews previously learned signs.



NPCs Interaction
: Introduces signs through conversations

Rewards Progression
: Gives rewards for learning progress



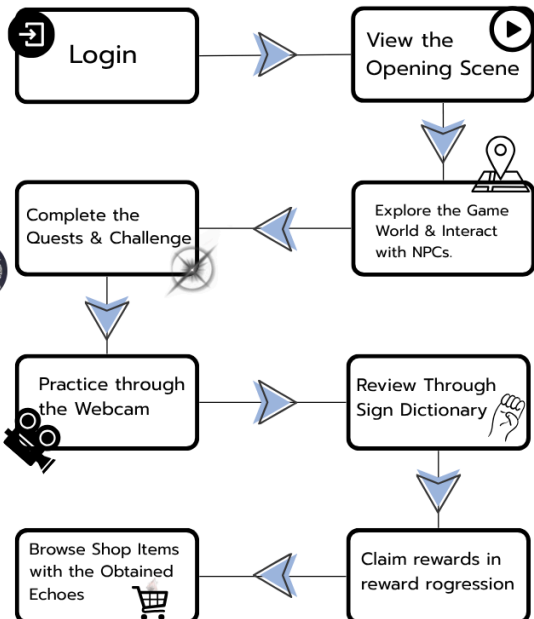
Webcam Practice
: Lets users practice signs.



Shop & Currency
: Buys items with earned currency



User Journey



```
graph TD; Login[Login] --> ViewScene[View the Opening Scene]; ViewScene --> ExploreGame[Explore the Game World & Interact with NPCs.]; ExploreGame --> CompleteQuests[Complete the Quests & Challenge]; CompleteQuests --> PracticeWebcam[Practice through the Webcam]; PracticeWebcam --> ReviewDict[Review Through Sign Dictionary]; ReviewDict --> BrowseShop[Browse Shop Items with the Obtained Echoes]; BrowseShop --> ClaimRewards[Claim rewards in reward regression];
```

Conclusion

Sonari: Path Unspoken is an interactive learning game that uses computer vision, machine learning, and the Observe-Understand-Apply model to help children and teenagers learn sign language while promoting inclusion, awareness, and confidence in communicating with the deaf community.

Technical Approach





Additional information (รายละเอียดเพิ่มเติม)

1. ต้องการพัฒนางานต่อยอดสู่ผู้ใช้งานจริงหลังเสร็จสิ้นกิจกรรม

สนใจ

2. โครงการต่อเนื่องจากโครงการเดิม

ไม่ต่อเนื่อง

3. รางวัลที่โครงการเคยได้รับจากเวทีอื่น

ไม่เคย

4. โครงการของท่านมีการทดลองเกี่ยวกับคนหรือสัตว์หรือไม่

ไม่ใช่

5. TRL

มีการสร้างแนวคิดทางเทคโนโลยี และ/หรือ การประยุกต์สูตรทางเทคโนโลยี

6. ระดับความพร้อมของระบบ (SRL)

2. มีการกำหนดปัญหา การเสนอแนวคิดในการพัฒนาหรือการแก้ปัญหา และคาดการณ์ผลกระทบที่อาจเกิดขึ้น และระบุผู้มีส่วนได้ส่วนเสียที่เกี่ยวข้องในโครงการ

7. SDGs

10. ช่วยลดความเหลื่อมล้ำ มุ่งลดความไม่เท่าเทียมทั้งภายในและระหว่างประเทศ